

A Weight-5 Apéry-like Binomial Sum, its Calabi–Yau Period, and Supercongruences

Xavier Callens

SocrateAI Laboratory

<https://github.com/xaviercallens/Mirror-Map-Sieve>

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Abstract

We study the binomial sum

$$S(n) = \sum_{k=0}^n \binom{n}{k}^4 \binom{n+k}{k},$$

an Apéry-like sequence of weight $(A, B) = (4, 1)$. We prove that $S(n)$ is the main diagonal of an explicit rational function in five variables, identifying the underlying geometry as the periods of a (compactified) Calabi–Yau 4-fold. By exact rational arithmetic, we extract the minimal holonomic recurrence—an order-5 linear recurrence with degree-9 polynomial coefficients—and determine the exact algebraic growth rate. We verify the Lian–Yau integrality of the associated mirror map up to degree $d = 16$. The base-case evaluations of the recurrence are certified by the Lean 4 kernel via the `decide` tactic, establishing axiom-free, computation-level verification of the arithmetic identities. Finally, we present computational evidence for two conjectural families of supercongruences: a Lucas congruence $S(mp + r) \equiv S(m)S(r) \pmod{p}$ for all primes $p \geq 5$, and a striking cubic supercongruence $S(p) \equiv S(1) \pmod{p^3}$ for all primes $p \geq 5$.

1 Introduction

Apéry’s celebrated proof of the irrationality of $\zeta(3)$ [1] relied on the sequence $\sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2$. Since then, classifying sequences of the form

$$S_{(A,B)}(n) = \sum_{k=0}^n \binom{n}{k}^A \binom{n+k}{k}^B \tag{1}$$

has been a fertile ground for discovering connections to modular forms, mirror symmetry, and Calabi–Yau varieties [2, 3, 4].

In this work, we present a detailed analysis of the case $(A, B) = (4, 1)$. Following the convention in (1), we write $S_{(4,1)}(n)$, or simply $S(n)$ when unambiguous:

$$S(n) = \sum_{k=0}^n \binom{n}{k}^4 \binom{n+k}{k}. \tag{2}$$

Remark 1. *The subscript notation “ S_{20} ” appearing in our code repository indexes this sequence within a broader catalog of (A, B) -pairs under investigation; it is not mathematically intrinsic to the sequence itself.*

The first few values of $S(n)$ for $n \geq 0$ are given in Table 1.

Our main contributions are:

n	0	1	2	3	4	5	6	7	8	9
$S(n)$	1	3	55	1155	29751	852753	26097499	840454275	28064517175	964417304253

Table 1: First 10 values of $S(n) = \sum_{k=0}^n \binom{n}{k}^4 \binom{n+k}{k}$.

1. A rigorous proof that $S(n)$ is a multivariate diagonal (Section 2).
2. Extraction of its minimal holonomic recurrence of order 5 and degree 9, with the explicit polynomial coefficients given in Appendix A (Section 3).
3. Axiom-free kernel-level computation of the base-case arithmetic identities in the Lean 4 proof assistant (Section 5).
4. Computational evidence for Lucas and cubic supercongruences (Section 6).
5. Verification of Lian–Yau integrality of the associated mirror map up to $d = 16$ (Section 4).

2 Diagonal Representation

A central result of our investigation is the exact diagonal representation of $S(n)$, which connects the sequence to multivariate algebraic geometry.

Theorem 2. *The sequence $S(n)$ is the main diagonal of the rational function*

$$R(x_1, x_2, x_3, x_4, x_5) = \frac{1}{\prod_{i=1}^5 (1 - x_i) - x_1 x_2 x_3 x_4}, \quad (3)$$

i.e., $S(n) = [x_1^n x_2^n x_3^n x_4^n x_5^n] R(x_1, \dots, x_5)$.

Proof. We expand R as a geometric series. Writing $Q = \prod_{i=1}^5 (1 - x_i)$ and $P = x_1 x_2 x_3 x_4$, we have

$$R = \frac{1}{Q - P} = \frac{1}{Q} \cdot \frac{1}{1 - P/Q} = \sum_{k=0}^{\infty} \frac{P^k}{Q^{k+1}} = \sum_{k=0}^{\infty} \frac{(x_1 x_2 x_3 x_4)^k}{\prod_{i=1}^5 (1 - x_i)^{k+1}}.$$

The series converges in a neighborhood of the origin. To extract the main diagonal coefficient $[x_1^n \cdots x_5^n]$, we extract the coefficient of each variable independently:

1. For x_i with $i \in \{1, 2, 3, 4\}$: we seek $[x_i^n]$ in $x_i^k / (1 - x_i)^{k+1}$, which equals $[x_i^{n-k}]$ in $(1 - x_i)^{-(k+1)}$. By the negative binomial expansion, this is $\binom{n-k+k}{k} = \binom{n}{k}$ when $0 \leq k \leq n$, and 0 otherwise. Taking the product over $i = 1, \dots, 4$ gives $\binom{n}{k}^4$.
2. For x_5 : the numerator contributes no factor of x_5 , so we simply extract $[x_5^n]$ from $(1 - x_5)^{-(k+1)} = \sum_j \binom{j+k}{k} x_5^j$, yielding $\binom{n+k}{k}$.

Multiplying and summing over k :

$$[x_1^n \cdots x_5^n] R = \sum_{k=0}^n \binom{n}{k}^4 \binom{n+k}{k} = S(n). \quad \square$$

Remark 3. *The zero locus of the denominator in (3), namely the affine hypersurface*

$$\prod_{i=1}^5 (1 - x_i) - x_1 x_2 x_3 x_4 = 0 \quad \subset \mathbb{A}^5,$$

is a (singular) Landau–Ginzburg model. The sequence $S(n)$ represents the periods of the appropriate toric compactification and crepant resolution of this hypersurface, yielding a smooth Calabi–Yau 4-fold. The periods of this variety satisfy the Picard–Fuchs differential equation whose recurrence form we extract in Section 3.

3 Minimal Recurrence and Asymptotics

By the Wilf–Zeilberger theory [7], the sequence $S(n)$ satisfies a linear recurrence with polynomial coefficients in n . To compute this recurrence explicitly, we employed two independent methods:

1. **$\mathbb{Q}$ -nullspace solver:** using exact integer arithmetic on the first 80 terms of $S(n)$, we constructed a large Hankel-like matrix and computed its rational nullspace, yielding candidate polynomial coefficients.
2. **Creative telescoping:** using SageMath’s implementation of Zeilberger’s algorithm for definite hypergeometric sums [7].

Both methods independently converged on the same result.

Theorem 4. *The sequence $S(n)$ satisfies a minimal linear recurrence of order 5:*

$$\sum_{j=0}^5 P_j(n) S(n+j) = 0, \quad (4)$$

where $P_0(n), \dots, P_5(n) \in \mathbb{Z}[n]$ are polynomials of degree 9, given explicitly in Appendix A.

The leading polynomial $P_5(n)$ is of particular geometric significance, as its roots (as a polynomial in n) correspond to the apparent singularities of the associated Picard–Fuchs differential operator. We have¹:

$$P_5(n) = 235032580722074992350169813838697598943355973 (n+1)^9 + \dots \quad (5)$$

with leading coefficient $\approx 2.35 \times 10^{44}$. The full polynomial is available in the repository’s `extracted_polynomials.json` file [10].

Exact asymptotics

The exponential growth rate of $S(n)$ is determined by a saddle-point analysis on the summand. Let t^* be the unique real root in $(0, 1)$ of

$$t^5 = (1-t)^4(1+t), \quad (6)$$

numerically $t^* \approx 0.56344$. The growth constant is then

$$\rho = \exp(\varphi(t^*)), \quad \varphi(t) = 4H(t) + (1+t)\ln(1+t) - t\ln t, \quad (7)$$

where $H(t) = -t\ln t - (1-t)\ln(1-t)$ is the binary entropy function. Numerically, $\rho \approx 43.0443$, and $S(n) \sim C \cdot \rho^n / n^2$ for an explicit constant $C > 0$.

4 Mirror Map Integrality

Following Lian and Yau [6], we extract the mirror map from the ratio of solutions of the associated Picard–Fuchs equation. Define

$$q(z) = z \cdot \exp\left(\frac{g(z)}{f(z)}\right),$$

where $f(z) = \sum_{n \geq 0} S(n) z^n$ is the holomorphic period and $g(z)$ is a logarithmic period constructed from f via the recurrence operator. The Lian–Yau integrality conjecture predicts that the Taylor coefficients q_d are integers.

¹The complete factorization is given in the ancillary data files of the repository [10].

Proposition 5 (Computational). *Using exact rational arithmetic (`fractions.Fraction` in Python), we verified that $q_d \in \mathbb{Z}$ for all $d = 1, 2, \dots, 16$.*

This provides strong computational evidence that $S(n)$ governs the mirror map of a genuine Calabi–Yau geometry, and that the instanton numbers (Gromov–Witten invariants) derived from this mirror map are integers.

5 Kernel-Verified Computation in Lean 4

To establish rigorous trust in our finite computations, we formalized the sequence definition and key arithmetic identities in the Lean 4 interactive theorem prover [9].

Remark 6 (Scope of the verification). *Our Lean 4 formalization uses the `decide` tactic, which instructs the kernel to evaluate finite integer expressions to normal form and verify equalities by definitional reduction. This provides axiom-free, kernel-certified computation of specific numerical identities—it is not a formal proof that the recurrence (4) holds for all $n \in \mathbb{N}$. A full formalization of the latter would require formalizing the rational creative telescoping certificate, which is a substantially larger undertaking beyond the scope of this work.*

The formalization, compiled with 0 `sorry` statements and 0 additional axioms, certifies:

1. The exact values $S(n)$ for $n = 0, 1, \dots, 7$.
2. The base-case identity at $n = 0$: the integer linear combination $\sum_{j=0}^5 P_j(0) \cdot S(j) = 0$, where $P_j(0)$ are the constant terms of the recurrence polynomials (reaching 46 digits each).
3. The base-case identity at $n = 1$, verified identically.

A representative excerpt of the proof:

```
def S20 (n : Nat) : Int :=
  ((Finset.range (n + 1)).sum
    (fun k => (Nat.choose n k)^4 * (Nat.choose (n + k) k)))

theorem s20_val_5 : S20 5 = 852753 := by decide

theorem recurrence_at_0 :
  (-541265085843..11840) * S20 0 +
  ... +
  (2047813495223..20000) * S20 5
  = 0 := by decide
```

The full Lean 4 source is available at [10].

6 Arithmetic Properties and Supercongruences

Apéry-like sequences are renowned for their deep arithmetic properties, including Lucas congruences and supercongruences modulo prime powers [4, 5]. We have conducted systematic computational experiments on $S(n)$, yielding the following conjectural results.

Conjecture 7 (Lucas congruence). *For every prime $p \geq 5$ and all non-negative integers m, r with $0 \leq r < p$,*

$$S(mp + r) \equiv S(m) \cdot S(r) \pmod{p}.$$

This has been verified computationally for all primes $p \leq 23$ and all m, r such that $mp + r \leq 79$.

Conjecture 8 (Apéry-style supercongruence). *For every prime $p \geq 3$,*

$$S(p-1) \equiv 1 \pmod{p^3}.$$

This has been verified for all primes $p \leq 23$. Note that this is the natural analogue of the Apéry supercongruence for the classical numbers $\sum_k \binom{n}{k}^2 \binom{n+k}{k}^2$.

Conjecture 9 (Cubic supercongruence at index p). *For every prime $p \geq 5$,*

$$S(p) \equiv S(1) = 3 \pmod{p^3}.$$

This striking congruence has been verified for all primes $p \leq 29$. It is notably stronger than the Lucas congruence (which would only guarantee $S(p) \equiv S(1) \pmod{p}$), and suggests that $S(n)$ may be connected to an underlying modular or automorphic form whose Hecke eigenvalues a_p satisfy particularly rigid arithmetic constraints.

The computational evidence for these congruences is summarized in Table 2.

p	$S(p) \bmod p^3$	$S(p-1) \bmod p^3$	Lucas mod p
5	3	1	✓
7	3	1	✓
11	3	1	✓
13	3	1	✓
17	3	1	✓
19	3	1	✓
23	3	1	✓
29	3	1	—

Table 2: Supercongruence data for $S(n)$. “Lucas mod p ” indicates whether $S(mp + r) \equiv S(m)S(r) \pmod{p}$ was verified for all accessible (m, r) pairs.

7 Conclusion

We have given a complete analysis of the weight- $(4, 1)$ Apéry-like binomial sum $S(n) = \sum_{k=0}^n \binom{n}{k}^4 \binom{n+k}{k}$: an explicit diagonal representation connecting it to a Calabi–Yau 4-fold, its minimal order-5 holonomic recurrence with degree-9 coefficients, exact asymptotics via saddle-point analysis, Lian–Yau mirror map integrality up to $d = 16$, and axiom-free kernel-level arithmetic verification in Lean 4.

Additionally, we have discovered three conjectural families of supercongruences (Conjectures 7–9), verified computationally for all accessible primes. The cubic supercongruence $S(p) \equiv 3 \pmod{p^3}$ is particularly striking and, to our knowledge, has not been previously observed for sequences of this type. Proving these conjectures, and understanding their connection to modular forms or p -adic cohomology, represents a natural direction for future work.

All computational artifacts—including the full recurrence polynomial coefficients, the Lean 4 proof source, verification scripts, and the first 80 terms of the sequence—are publicly available at [10].

A Recurrence Polynomial Coefficients

The minimal recurrence $\sum_{j=0}^5 P_j(n) S(n+j) = 0$ has polynomial coefficients $P_j(n) \in \mathbb{Z}[n]$ of degree 9. Due to the size of the integer coefficients (up to 46 digits), we present them in compact notation. For each polynomial, we write $P_j(n) = \sum_{i=0}^9 c_{j,i} n^i$ and list the coefficient vector $(c_{j,0}, c_{j,1}, \dots, c_{j,9})$.

$P_0(n)$: The constant term is $P_0(0) = -5412650858431135013634958175726842170573378411840$.

$P_5(n)$: The constant term is $P_5(0) = 20478134952232355172884134183653971676016433020000$, and the leading coefficient is $c_{5,9} = 235032580722074992350169813838697598943355973$.

The complete set of all 60 integer coefficients (6 polynomials $\times$ 10 coefficients each) is provided in machine-readable form in the file `extracted_polynomials.json` in the repository [10]. These coefficients were independently verified by SageMath’s creative telescoping and by exact $\mathbb{Q}$ -nullspace computation on 80 terms.

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